How to Conduct Inference with Spatial Dependence in Stata

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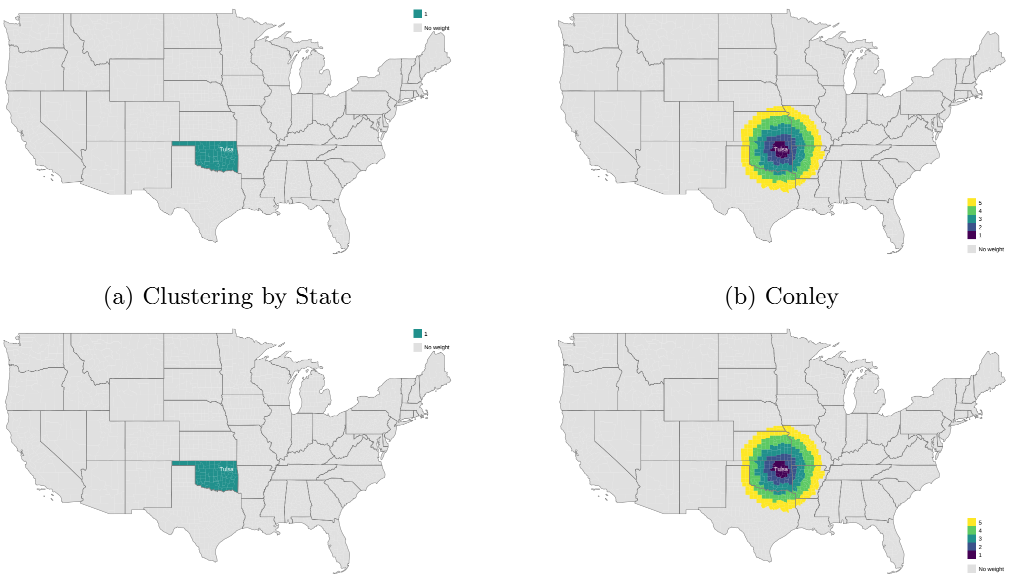
Abstract. In this paper...

Keywords: spatial dependence, inference.

1 Introduction

DellaVigna et al. (2025)

Conley (1999)



(a) Clustering by State (b) Conley

(c) SCPC (d) TMO

Figure 1: Spatial Standard Error Procedures

2 Methods

Consider an equation of the form . . .

Moulton (1986)

Müller and Watson (2022)

3 Syntax

4 Examples

```
. use ../example/county_differences
. qui ds fips stfips PIN_persincpc_d EDU_college_d, not
. local ylist `r(varlist)'
.
```

```
. tmo, cmd(reg PIN_persincpc_d EDU_college_d i.stfips, r) x(EDU_college_d) ylist(`ylist´) i(fips)
Linear Regression with TMO                                    Number of obs =  3,060
                                                              R-squared     = 0.3434
                                                              Adj R-squared = 0.3434
                                                              Root MSE      = 0.1409
```

PIN_persinc~d	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
EDU_college_d	.1745893	.0197081	8.86	0.000	.1359466	.213232

Optimal threshold	0.389
% of off-diag in SE est.	0.497
% >= threshold (excl. clusters/Conley)	0.497
# outcomes	90.000
Degrees of freedom	59.986

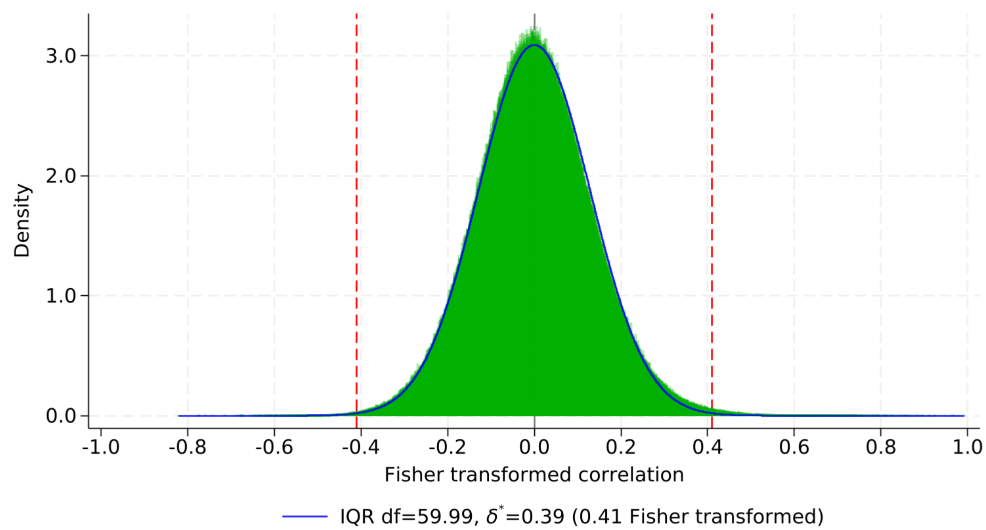


Figure 3: Correlations between residuals across US counties

5 Conclusion

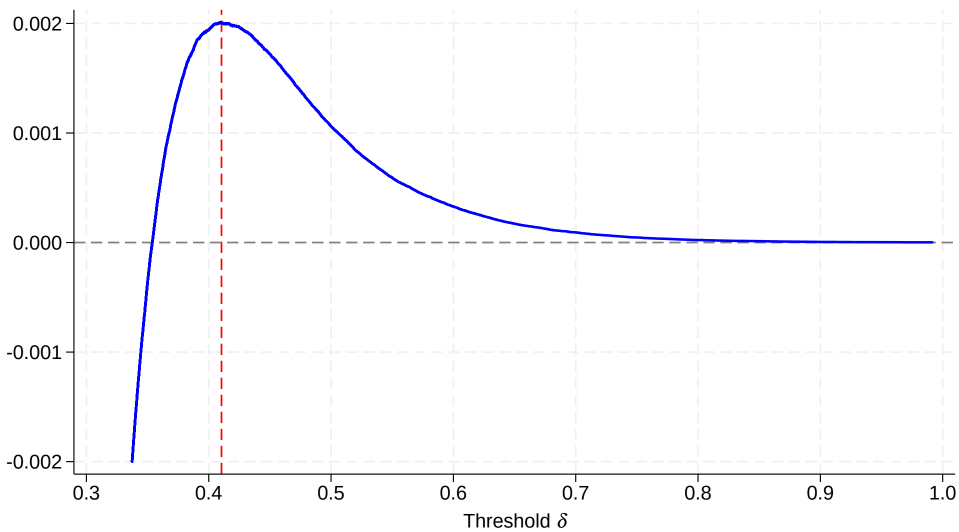


Figure 4: Optimal Thresholding

Acknowledgments

We are grateful to . . .

6 References

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